PROGRESS

Status values: **Planned** → **Drafted** → **Validated** → **Completed**.

- *Drafted*: written and structurally checked (15 sections, relative links resolve).
- *Validated*: independent review checked claims against primary sources and ran or typechecked the code.
- *Completed*: Validated and all external URLs returned HTTP 200.

Honest state (generated 2026-10-22): every lesson below passed an independent validation pass (claims checked against primary sources, code executed/typechecked in a sandbox, ~440 corrections applied; see REVIEW-NOTES.md and validation-reports/). **Completed** = validated and no external link remains marked *(unverified)*. **Validated** = validated, but at least one link could not be checked automatically. No human expert audit has been done. Each PDF sits next to its Markdown file.

Learner checkboxes: tick only when the knowledge checkpoint is met without notes.

Phase 0 — Web and JavaScript foundations (`00-web-and-javascript-foundations/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-how-the-browser-works	Completed	✓	[ ]	[ ]	[ ]
02	02-html-parsing-dom-cssom-render-tree	Completed	✓	[ ]	[ ]	[ ]
03	03-browser-rendering-pipeline	Completed	✓	[ ]	[ ]	[ ]
04	04-dom-structure-and-mutation	Completed	✓	[ ]	[ ]	[ ]
05	05-execution-contexts-and-call-stack	Completed	✓	[ ]	[ ]	[ ]
06	06-lexical-scope-and-closures	Completed	✓	[ ]	[ ]	[ ]
07	07-functions-and-referential-identity	Completed	✓	[ ]	[ ]	[ ]
08	08-objects-references-immutability	Validated	✓	[ ]	[ ]	[ ]
09	09-prototypes-and-object-model	Validated	✓	[ ]	[ ]	[ ]
10	10-modules-and-dependency-graphs	Validated	✓	[ ]	[ ]	[ ]
11	11-event-loop-tasks-microtasks	Validated	✓	[ ]	[ ]	[ ]
12	12-promises-and-async-await	Completed	✓	[ ]	[ ]	[ ]
13	13-browser-events-and-propagation	Validated	✓	[ ]	[ ]	[ ]
14	14-web-apis-and-network-requests	Validated	✓	[ ]	[ ]	[ ]
15	15-http-caching-and-navigation	Validated	✓	[ ]	[ ]	[ ]
16	16-memory-management-and-gc	Validated	✓	[ ]	[ ]	[ ]
17	17-javascript-performance-fundamentals	Validated	✓	[ ]	[ ]	[ ]

#	File	Status	PDF	Read	Exercises	Checkpoint
18	18-typescript-fundamentals-and-narrowing	Validated	✓	[ ]	[ ]	[ ]
19	19-typescript-generics-and-utility-types	Validated	✓	[ ]	[ ]	[ ]
20	20-tooling-transpilation-and-bundling	Validated	✓	[ ]	[ ]	[ ]
21	21-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 1 — React fundamentals (`01-react-fundamentals/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-what-is-react	Validated	✓	[ ]	[ ]	[ ]
02	02-declarative-vs-imperative-ui	Completed	✓	[ ]	[ ]	[ ]
03	03-ui-as-a-function-of-state	Validated	✓	[ ]	[ ]	[ ]
04	04-elements-components-instances	Validated	✓	[ ]	[ ]	[ ]
05	05-jsx-explained	Validated	✓	[ ]	[ ]	[ ]
06	06-props-and-one-way-data-flow	Validated	✓	[ ]	[ ]	[ ]
07	07-composition-and-children	Validated	✓	[ ]	[ ]	[ ]
08	08-conditional-rendering	Validated	✓	[ ]	[ ]	[ ]
09	09-lists-keys-and-identity	Completed	✓	[ ]	[ ]	[ ]
10	10-event-handling	Completed	✓	[ ]	[ ]	[ ]
11	11-controlled-and-uncontrolled-inputs	Completed	✓	[ ]	[ ]	[ ]
12	12-component-purity	Completed	✓	[ ]	[ ]	[ ]
13	13-rendering-mental-model	Completed	✓	[ ]	[ ]	[ ]
14	14-state-as-a-snapshot	Completed	✓	[ ]	[ ]	[ ]
15	15-update-queues-and-batching	Completed	✓	[ ]	[ ]	[ ]
16	16-state-preservation-and-reset	Completed	✓	[ ]	[ ]	[ ]
17	17-lifecycle-concepts	Completed	✓	[ ]	[ ]	[ ]
18	18-react-devtools	Completed	✓	[ ]	[ ]	[ ]
19	19-project-setup-vite-typescript	Completed	✓	[ ]	[ ]	[ ]
20	20-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 2 — Rendering, reconciliation, DOM, internals (`02-react-rendering-and-internals/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-what-triggers-a-render	Completed	✓	[ ]	[ ]	[ ]

#	File	Status	PDF	Read	Exercises	Checkpoint
02	02-render-reconcile-commit-paint	Validated	✓	[ ]	[ ]	[ ]
03	03-initial-mount-and-updates	Completed	✓	[ ]	[ ]	[ ]
04	04-virtual-dom-what-it-means	Validated	✓	[ ]	[ ]	[ ]
05	05-elements-vs-dom-nodes	Completed	✓	[ ]	[ ]	[ ]
06	06-reconciliation-fundamentals	Completed	✓	[ ]	[ ]	[ ]
07	07-identity-keys-and-reconciliation	Completed	✓	[ ]	[ ]	[ ]
08	08-why-fiber	Validated	✓	[ ]	[ ]	[ ]
09	09-fiber-nodes-and-units-of-work	Completed	✓	[ ]	[ ]	[ ]
10	10-current-and-work-in-progress-trees	Completed	✓	[ ]	[ ]	[ ]
11	11-render-phase-and-commit-phase	Completed	✓	[ ]	[ ]	[ ]
12	12-why-rendering-must-be-pure	Completed	✓	[ ]	[ ]	[ ]
13	13-scheduler-and-priorities	Completed	✓	[ ]	[ ]	[ ]
14	14-lanes	Completed	✓	[ ]	[ ]	[ ]
15	15-interruptible-restartable-abandoned-rendering	Validated	✓	[ ]	[ ]	[ ]
16	16-concurrent-react-misconceptions	Completed	✓	[ ]	[ ]	[ ]
17	17-hooks-and-fiber-state	Completed	✓	[ ]	[ ]	[ ]
18	18-effects-and-their-timing	Completed	✓	[ ]	[ ]	[ ]
19	19-layout-vs-passive-effects	Completed	✓	[ ]	[ ]	[ ]
20	20-commit-and-dom-mutation	Completed	✓	[ ]	[ ]	[ ]
21	21-browser-paint-and-react-updates	Completed	✓	[ ]	[ ]	[ ]
22	22-automatic-batching	Completed	✓	[ ]	[ ]	[ ]
23	23-transitions-and-deferred-rendering	Completed	✓	[ ]	[ ]	[ ]
24	24-suspense-conceptual-internals	Completed	✓	[ ]	[ ]	[ ]
25	25-hydration-and-selective-hydration	Completed	✓	[ ]	[ ]	[ ]
26	26-ssr-and-client-rendering	Completed	✓	[ ]	[ ]	[ ]
27	27-react-server-components-intro	Completed	✓	[ ]	[ ]	[ ]
28	28-strict-mode-behavior	Completed	✓	[ ]	[ ]	[ ]
29	29-reasoning-about-unnecessary-renders	Completed	✓	[ ]	[ ]	[ ]
30	30-react-devtools-profiler	Validated	✓	[ ]	[ ]	[ ]
31	31-reading-react-source	Validated	✓	[ ]	[ ]	[ ]
32	32-scenario-a-parent-rerender	Completed	✓	[ ]	[ ]	[ ]
33	33-scenario-b-text-update	Completed	✓	[ ]	[ ]	[ ]
34	34-scenario-c-list-reorder	Completed	✓	[ ]	[ ]	[ ]
35	35-scenario-d-transition-vs-urgent	Completed	✓	[ ]	[ ]	[ ]

#	File	Status	PDF	Read	Exercises	Checkpoint
36	36-scenario-e-suspend	Completed	✓	[ ]	[ ]	[ ]
37	37-scenario-f-hydration-mismatch	Completed	✓	[ ]	[ ]	[ ]
38	38-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 3 — State management and Hooks (`03-react-state-and-hooks/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-usestate-internals-and-update-semantics	Completed	✓	[ ]	[ ]	[ ]
02	02-functional-state-updates	Completed	✓	[ ]	[ ]	[ ]
03	03-usereducer-and-reducer-architecture	Completed	✓	[ ]	[ ]	[ ]
04	04-useeffect-synchronization	Completed	✓	[ ]	[ ]	[ ]
05	05-effect-dependencies-and-stale-closures	Completed	✓	[ ]	[ ]	[ ]
06	06-effect-cleanup	Completed	✓	[ ]	[ ]	[ ]
07	07-you-might-not-need-an-effect	Completed	✓	[ ]	[ ]	[ ]
08	08-uselayouteffect	Completed	✓	[ ]	[ ]	[ ]
09	09-useinsertioneffect	Completed	✓	[ ]	[ ]	[ ]
10	10-useref-and-mutable-references	Validated	✓	[ ]	[ ]	[ ]
11	11-usememo-and-usecallback	Completed	✓	[ ]	[ ]	[ ]
12	12-react-memo	Completed	✓	[ ]	[ ]	[ ]
13	13-usecontext-and-propagation	Completed	✓	[ ]	[ ]	[ ]
14	14-context-performance	Completed	✓	[ ]	[ ]	[ ]
15	15-custom-hooks	Completed	✓	[ ]	[ ]	[ ]
16	16-rules-of-hooks-and-composition	Completed	✓	[ ]	[ ]	[ ]
17	17-derived-state	Completed	✓	[ ]	[ ]	[ ]
18	18-state-colocation-and-lifting	Completed	✓	[ ]	[ ]	[ ]
19	19-ui-server-url-and-shared-state	Completed	✓	[ ]	[ ]	[ ]
20	20-usesyncexternalstore-and-external-stores	Validated	✓	[ ]	[ ]	[ ]
21	21-forms-and-complex-state	Completed	✓	[ ]	[ ]	[ ]
22	22-modern-hooks-and-apis	Completed	✓	[ ]	[ ]	[ ]
23	23-hooks-anti-patterns	Completed	✓	[ ]	[ ]	[ ]
24	24-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 4 — React architecture (`04-react-architecture/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-architecture-fundamentals	Validated	✓	[ ]	[ ]	[ ]
02	02-separation-of-concerns	Validated	✓	[ ]	[ ]	[ ]
03	03-component-responsibility	Completed	✓	[ ]	[ ]	[ ]
04	04-presentational-vs-container	Validated	✓	[ ]	[ ]	[ ]
05	05-composition-and-dependency-direction	Completed	✓	[ ]	[ ]	[ ]
06	06-feature-vs-layer-structure	Completed	✓	[ ]	[ ]	[ ]
07	07-modular-architecture	Validated	✓	[ ]	[ ]	[ ]
08	08-domain-driven-frontend	Validated	✓	[ ]	[ ]	[ ]
09	09-feature-sliced-design	Validated	✓	[ ]	[ ]	[ ]
10	10-clean-and-hexagonal-architecture	Validated	✓	[ ]	[ ]	[ ]
11	11-dependency-inversion	Validated	✓	[ ]	[ ]	[ ]
12	12-domain-logic-vs-ui-logic	Validated	✓	[ ]	[ ]	[ ]
13	13-application-services-and-adapters	Validated	✓	[ ]	[ ]	[ ]
14	14-api-boundaries-and-data-access	Completed	✓	[ ]	[ ]	[ ]
15	15-state-ownership	Completed	✓	[ ]	[ ]	[ ]
16	16-server-vs-client-state	Completed	✓	[ ]	[ ]	[ ]
17	17-data-flow-architecture	Completed	✓	[ ]	[ ]	[ ]
18	18-routing-architecture	Completed	✓	[ ]	[ ]	[ ]
19	19-error-boundaries	Completed	✓	[ ]	[ ]	[ ]
20	20-design-systems	Validated	✓	[ ]	[ ]	[ ]
21	21-component-libraries	Validated	✓	[ ]	[ ]	[ ]
22	22-monorepos-and-shared-packages	Validated	✓	[ ]	[ ]	[ ]
23	23-multi-team-ownership	Validated	✓	[ ]	[ ]	[ ]
24	24-refactoring-legacy-react	Validated	✓	[ ]	[ ]	[ ]
25	25-architectural-fitness-and-enforcement	Validated	✓	[ ]	[ ]	[ ]
26	26-phase-assessment	Completed	✓	[ ]	[ ]	[ ]
27	27-styling-architecture	Validated	✓	[ ]	[ ]	[ ]

Phase 5 — Design patterns (`05-design-patterns/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-solid-in-react	Validated	✓	[ ]	[ ]	[ ]
02	02-dry-kiss-yagni	Validated	✓	[ ]	[ ]	[ ]
03	03-composition-over-inheritance	Validated	✓	[ ]	[ ]	[ ]

#	File	Status	PDF	Read	Exercises	Checkpoint
04	04-container-presentational	Validated	✓	[ ]	[ ]	[ ]
05	05-compound-components	Validated	✓	[ ]	[ ]	[ ]
06	06-custom-hook-patterns	Validated	✓	[ ]	[ ]	[ ]
07	07-headless-components	Validated	✓	[ ]	[ ]	[ ]
08	08-provider-pattern	Validated	✓	[ ]	[ ]	[ ]
09	09-higher-order-components	Validated	✓	[ ]	[ ]	[ ]
10	10-render-props	Validated	✓	[ ]	[ ]	[ ]
11	11-controlled-component-pattern	Validated	✓	[ ]	[ ]	[ ]
12	12-reducer-pattern	Validated	✓	[ ]	[ ]	[ ]
13	13-state-machine-pattern	Completed	✓	[ ]	[ ]	[ ]
14	14-observer-and-subscription	Validated	✓	[ ]	[ ]	[ ]
15	15-adapter-strategy-factory	Validated	✓	[ ]	[ ]	[ ]
16	16-dependency-injection	Completed	✓	[ ]	[ ]	[ ]
17	17-facade-and-command	Completed	✓	[ ]	[ ]	[ ]
18	18-behavior-vs-presentation	Completed	✓	[ ]	[ ]	[ ]
19	19-polymorphic-components	Validated	✓	[ ]	[ ]	[ ]
20	20-slot-patterns	Validated	✓	[ ]	[ ]	[ ]
21	21-error-handling-patterns	Validated	✓	[ ]	[ ]	[ ]
22	22-patterns-for-reusable-libraries	Validated	✓	[ ]	[ ]	[ ]
23	23-patterns-made-less-necessary	Validated	✓	[ ]	[ ]	[ ]
24	24-oop-patterns-in-functional-react	Validated	✓	[ ]	[ ]	[ ]
25	25-comparative-refactoring-labs	Validated	✓	[ ]	[ ]	[ ]
26	26-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 6 — Data fetching and state ecosystems (`06-data-fetching-and-state-ecosystems/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-fetch-lifecycle	Validated	✓	[ ]	[ ]	[ ]
02	02-cancellation-and-race-conditions	Validated	✓	[ ]	[ ]	[ ]
03	03-loading-and-error-states	Validated	✓	[ ]	[ ]	[ ]
04	04-api-normalization	Validated	✓	[ ]	[ ]	[ ]
05	05-optimistic-updates	Validated	✓	[ ]	[ ]	[ ]
06	06-cache-consistency-and-invalidation	Validated	✓	[ ]	[ ]	[ ]

#	File	Status	PDF	Read	Exercises	Checkpoint
07	07-server-state-management	Validated	✓	[ ]	[ ]	[ ]
08	08-tanstack-query	Completed	✓	[ ]	[ ]	[ ]
09	09-redux-toolkit	Completed	✓	[ ]	[ ]	[ ]
10	10-zustand	Completed	✓	[ ]	[ ]	[ ]
11	11-context-vs-external-stores	Completed	✓	[ ]	[ ]	[ ]
12	12-state-machines-and-xstate	Validated	✓	[ ]	[ ]	[ ]
13	13-suspense-data-loading	Completed	✓	[ ]	[ ]	[ ]
14	14-server-side-data-loading	Completed	✓	[ ]	[ ]	[ ]
15	15-error-recovery-and-retries	Completed	✓	[ ]	[ ]	[ ]
16	16-offline-considerations	Completed	✓	[ ]	[ ]	[ ]
17	17-real-time-websockets-subscriptions	Validated	✓	[ ]	[ ]	[ ]
18	18-url-state-management	Completed	✓	[ ]	[ ]	[ ]
19	19-technology-selection-matrix	Completed	✓	[ ]	[ ]	[ ]
20	20-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 7 — Performance engineering (`07-performance/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-measure-before-optimizing	Validated	✓	[ ]	[ ]	[ ]
02	02-profiler-workflows	Validated	✓	[ ]	[ ]	[ ]
03	03-why-components-rerender	Validated	✓	[ ]	[ ]	[ ]
04	04-memoization-trade-offs	Validated	✓	[ ]	[ ]	[ ]
05	05-referential-equality	Validated	✓	[ ]	[ ]	[ ]
06	06-context-propagation-cost	Validated	✓	[ ]	[ ]	[ ]
07	07-state-colocation-for-perf	Completed	✓	[ ]	[ ]	[ ]
08	08-rendering-large-collections	Completed	✓	[ ]	[ ]	[ ]
09	09-windowing-and-virtualization	Completed	✓	[ ]	[ ]	[ ]
10	10-expensive-computations	Completed	✓	[ ]	[ ]	[ ]
11	11-scheduling-and-perceived-responsiveness	Completed	✓	[ ]	[ ]	[ ]
12	12-usetransition-in-practice	Completed	✓	[ ]	[ ]	[ ]
13	13-usedeferredvalue-in-practice	Completed	✓	[ ]	[ ]	[ ]
14	14-debounce-and-throttle	Completed	✓	[ ]	[ ]	[ ]
15	15-code-splitting-and-lazy-loading	Validated	✓	[ ]	[ ]	[ ]
16	16-bundle-analysis	Completed	✓	[ ]	[ ]	[ ]

#	File	Status	PDF	Read	Exercises	Checkpoint
17	17-network-waterfalls	Completed	✓	[ ]	[ ]	[ ]
18	18-react-vs-browser-bottlenecks	Completed	✓	[ ]	[ ]	[ ]
19	19-memory-leaks	Completed	✓	[ ]	[ ]	[ ]
20	20-long-tasks-and-main-thread	Completed	✓	[ ]	[ ]	[ ]
21	21-web-vitals-and-budgets	Completed	✓	[ ]	[ ]	[ ]
22	22-production-performance-investigations	Completed	✓	[ ]	[ ]	[ ]
23	23-counterproductive-optimizations	Completed	✓	[ ]	[ ]	[ ]
24	24-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 8 — Testing and quality (`08-testing-and-quality/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-test-strategy-and-pyramid	Validated	✓	[ ]	[ ]	[ ]
02	02-unit-testing-with-vitest	Completed	✓	[ ]	[ ]	[ ]
03	03-react-testing-library	Validated	✓	[ ]	[ ]	[ ]
04	04-testing-async-updates	Validated	✓	[ ]	[ ]	[ ]
05	05-testing-custom-hooks	Validated	✓	[ ]	[ ]	[ ]
06	06-mocking-strategies-and-msw	Validated	✓	[ ]	[ ]	[ ]
07	07-testing-api-interactions	Validated	✓	[ ]	[ ]	[ ]
08	08-integration-testing	Validated	✓	[ ]	[ ]	[ ]
09	09-e2e-with-playwright	Validated	✓	[ ]	[ ]	[ ]
10	10-accessibility-testing	Completed	✓	[ ]	[ ]	[ ]
11	11-component-contract-testing	Completed	✓	[ ]	[ ]	[ ]
12	12-visual-regression	Completed	✓	[ ]	[ ]	[ ]
13	13-typescript-correctness	Completed	✓	[ ]	[ ]	[ ]
14	14-linting-and-static-analysis	Completed	✓	[ ]	[ ]	[ ]
15	15-ci-checks	Completed	✓	[ ]	[ ]	[ ]
16	16-avoiding-brittle-tests	Completed	✓	[ ]	[ ]	[ ]
17	17-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 9 — Advanced rendering and ecosystem (`09-advanced-rendering-and-ecosystem/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-spa-ssr-ssg-streaming	Completed	✓	[ ]	[ ]	[ ]
02	02-hydration-internals-deep-dive	Completed	✓	[ ]	[ ]	[ ]
03	03-server-components-in-depth	Validated	✓	[ ]	[ ]	[ ]
04	04-client-server-boundaries-and-serialization	Validated	✓	[ ]	[ ]	[ ]
05	05-suspense-architecture	Completed	✓	[ ]	[ ]	[ ]
06	06-concurrent-rendering-in-practice	Completed	✓	[ ]	[ ]	[ ]
07	07-modern-react-apis	Validated	✓	[ ]	[ ]	[ ]
08	08-react-compiler	Validated	✓	[ ]	[ ]	[ ]
09	09-framework-architecture	Completed	✓	[ ]	[ ]	[ ]
10	10-routing-and-nested-layouts	Completed	✓	[ ]	[ ]	[ ]
11	11-progressive-enhancement	Completed	✓	[ ]	[ ]	[ ]
12	12-auth-boundaries	Completed	✓	[ ]	[ ]	[ ]
13	13-forms-actions-and-mutations	Completed	✓	[ ]	[ ]	[ ]
14	14-accessibility-and-seo	Completed	✓	[ ]	[ ]	[ ]
15	15-react-api-changes-across-versions	Completed	✓	[ ]	[ ]	[ ]
16	16-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 10 — Production engineering (`10-production-engineering/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-repository-organization	Validated	✓	[ ]	[ ]	[ ]
02	02-deployment-workflows	Validated	✓	[ ]	[ ]	[ ]
03	03-environment-configuration	Validated	✓	[ ]	[ ]	[ ]
04	04-observability-and-logging	Validated	✓	[ ]	[ ]	[ ]
05	05-error-monitoring	Validated	✓	[ ]	[ ]	[ ]
06	06-security-xss-csrf-csp	Validated	✓	[ ]	[ ]	[ ]
07	07-dependency-security	Validated	✓	[ ]	[ ]	[ ]
08	08-accessibility-as-architecture	Validated	✓	[ ]	[ ]	[ ]
09	09-internationalization	Validated	✓	[ ]	[ ]	[ ]
10	10-resilience-and-degradation	Validated	✓	[ ]	[ ]	[ ]
11	11-design-system-governance	Validated	✓	[ ]	[ ]	[ ]
12	12-backward-compatibility	Validated	✓	[ ]	[ ]	[ ]
13	13-application-migrations	Validated	✓	[ ]	[ ]	[ ]
14	14-multi-team-collaboration	Validated	✓	[ ]	[ ]	[ ]

#	File	Status	PDF	Read	Exercises	Checkpoint
15	15-technical-debt	Validated	✓	[ ]	[ ]	[ ]
16	16-architecture-decision-records	Validated	✓	[ ]	[ ]	[ ]
17	17-documentation	Validated	✓	[ ]	[ ]	[ ]
18	18-continuous-delivery	Validated	✓	[ ]	[ ]	[ ]
19	19-incident-debugging	Validated	✓	[ ]	[ ]	[ ]
20	20-phase-assessment	Completed	✓	[ ]	[ ]	[ ]

Phase 11 — Senior and architect mastery (`11-senior-and-architect-mastery/`)

#	File	Status	PDF	Read	Exercises	Checkpoint
01	01-trade-off-analysis	Validated	✓	[ ]	[ ]	[ ]
02	02-technical-design-documents	Validated	✓	[ ]	[ ]	[ ]
03	03-large-scale-refactoring	Validated	✓	[ ]	[ ]	[ ]
04	04-choosing-abstraction-level	Validated	✓	[ ]	[ ]	[ ]
05	05-application-boundaries	Validated	✓	[ ]	[ ]	[ ]
06	06-cross-cutting-concerns	Validated	✓	[ ]	[ ]	[ ]
07	07-design-system-architecture	Validated	✓	[ ]	[ ]	[ ]
08	08-state-architecture-at-scale	Validated	✓	[ ]	[ ]	[ ]
09	09-performance-budgets-at-scale	Validated	✓	[ ]	[ ]	[ ]
10	10-designing-for-testability	Completed	✓	[ ]	[ ]	[ ]
11	11-designing-for-accessibility	Validated	✓	[ ]	[ ]	[ ]
12	12-microfrontends	Validated	✓	[ ]	[ ]	[ ]
13	13-monorepo-vs-multirepo	Validated	✓	[ ]	[ ]	[ ]
14	14-build-vs-buy	Validated	✓	[ ]	[ ]	[ ]
15	15-framework-selection	Validated	✓	[ ]	[ ]	[ ]
16	16-supporting-multiple-teams	Validated	✓	[ ]	[ ]	[ ]
17	17-code-review-and-standards	Validated	✓	[ ]	[ ]	[ ]
18	18-technical-leadership	Validated	✓	[ ]	[ ]	[ ]
19	19-debugging-with-incomplete-information	Validated	✓	[ ]	[ ]	[ ]
20	20-evaluating-emerging-features	Validated	✓	[ ]	[ ]	[ ]
21	21-reading-framework-source	Validated	✓	[ ]	[ ]	[ ]
22	22-long-term-evolution	Validated	✓	[ ]	[ ]	[ ]
23	23-system-design-interviews	Validated	✓	[ ]	[ ]	[ ]
24	24-final-assessment	Validated	✓	[ ]	[ ]	[ ]

Projects (`projects/`)

#	File	Status	Done
01	01-interactive-fundamentals-app	Completed	[ ]
02	02-task-manager-state-ownership	Completed	[ ]
03	03-multi-page-dashboard	Completed	[ ]
04	04-complex-form-system	Completed	[ ]
05	05-data-heavy-management-ui	Completed	[ ]
06	06-ecommerce-server-state-caching	Completed	[ ]
07	07-reusable-component-library	Completed	[ ]
08	08-enterprise-modular-app	Completed	[ ]
09	09-ssr-streaming-app	Completed	[ ]
10	10-capstone	Validated	[ ]
11	11-broken-app-diagnose-and-refactor	Validated	[ ]

Total content files: 288.

Assessment passing rule

≥80% overall **and** ≥70% in each category (explanation, implementation, debugging, reasoning) before advancing; see each phase's `*-phase-assessment.md`. The final assessment is `11-senior-and-architect-mastery/24-final-assessment.md`.